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Systematic Literature Review in the Development of Datasets and Fact Verification Models for Indonesian Language

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***Abstract*—This study explores the major challenges in fact verification for the Indonesian language due to gaps in managing dynamic information, uncertainties from linguistic variations, and the proliferation of online narratives. Most text classification research for fact verification relies on English-language datasets, which may not be suitable for the Indonesian context. Through a Systematic Literature Review (SLR) approach, this research analyzes the development of datasets and models for Indonesian-language fact verification tasks. Utilizing the PICOC framework, this research conducted a systematic literature review to explore the development of datasets and models for Indonesian-language fact verification tasks. The main objectives of this research are to identify existing research shortcomings, evaluate the performance of current fact verification models, and propose recommendations for future research. These findings highlight the lack of research specifically addressing fact verification in Indonesia, as well as the lack of datasets and models specifically designed for the fact verification context. Existing models often perform poorly in terms of precision, recall, and F1 score, with average scores below 70%, indicating difficulty in identifying relevant evidence and consistently verifying facts. The key findings indicate a lack of research specifically addressing fact verification in Indonesia, as well as limited datasets and models specifically designed for fact verification. Additionally, most fact-verification tasks do not cover specific topics, except those related to politics. Common performance metrics used in this analysis include accuracy, precision, recall, F1-Score, and Dataset Score.**

Keywords—fact verification, dataset, model, indonesian-language

I. INTRODUCTION

The Indonesian language fact-checking mechanism for digital content has become a primary need, relied upon to filter the large amount of information circulating in online media [1], [2], [3], [4] due to the increasing prevalence of fake news in Indonesia [5]. The development of social media platforms and online news sources goes hand in hand with the importance of providing information that has authenticity and data accuracy [6]. Ensuring that information is true news (not fake news) is a challenging task in combating the spread of fake news, which has a negative impact if news consumers believe it . In an era where information can quickly go viral, the vulnerability to disinformation is increasing, creating the need for reliable and effective fact verification mechanisms in the digital environment, by providing evidence for news claims [7], [8]. However, fact verification has now become a primary need along with the easy dissemination of information. Fig. 1 shows the statistics of fake news categories spread in Indonesia, sourced from the Ministry of Communication and Information Technology (KOMINFO) Indonesia [9]. Health, government, and politics are the most targeted topics for fake news.

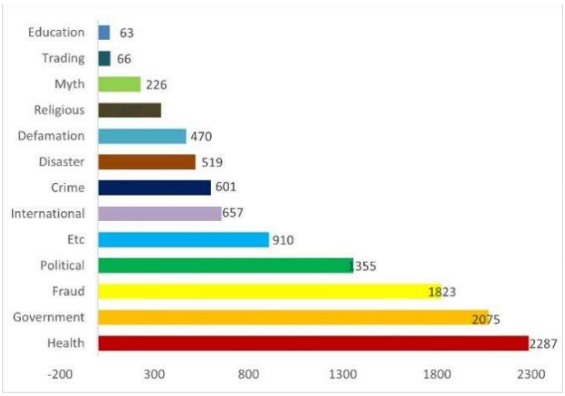


Fig. 1. Statistics on the Distribution of Fake News in Indonesia

These topics pose significant challenges for fact verification in managing information in online media. Fake news contains false or misleading information [10], hereby causing social tension and affecting political stability and democracy. The phenomenon of spreading fake news goes hand in hand with technological advances that facilitate the dissemination of information [11], [12], [13]. Uncertainty arises from linguistic variation and the evolution of online narratives, which can spread inaccurate information. This gap is exacerbated by fact verification methods' limitations in reacting to change and complex language. An innovative solution to dealing with fake news is to create new datasets [14], [15], [16], [17] and apply deep learning technology [18]. Christodoulopoulos et al. developed the FEVEROUS dataset [19] for fact verification.

Besides FEVEROUS, EX-FEVER [20] and FACTIFY-5WQA [21] were also developed to create fact verification

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datasets in English. FEVEROUS provides claims and supporting facts, while EX-FEVER and FACTIFY-5WQA are used to train claim detection and verification models. Algorithms from the Large Language Models (LLMs), such as BERT [22], RoBERTa [23], and GPT [24], are considered capable of performing fact verification tasks [25], [26]. These algorithms are necessary to understand the context and language of the claim. Synergy between dataset and algorithm development is critical for optimal results. Good datasets provide training material for algorithms, while innovative algorithms maximize the potential of datasets. However, in Indonesia, fact verification datasets and models equivalent to English ones are still limited, especially in providing evidence for each claim and fitting models to Indonesian fact verification datasets. To address this gap, SLR was chosen because this study aims to systematically analyze the existing literature on the development of datasets and fact verification models, particularly those related to the Indonesian language. Through SLR, this study ensures that the analyzed literature meets the established inclusion and exclusion criteria, avoids bias, and provides a comprehensive view of the shortcomings and challenges in this field. In general, the main focus of this literature study is related to databases and fact verification models to reduce the spread of fake news in Indonesia. Specifically, the objectives of this research are:

- Conduct a systematic review of available literature to collect and analyze the latest research on dataset development and fact verification models.
- Analyze and evaluate deficiencies in current fact verification models by considering the performance evaluation results of datasets and fact verification models.
- Propose recommendations or guidelines for improving the development of datasets and fact verification models in the future.

In this paper, the discussion is divided into several parts. Section 2 discusses literature review on the topic of fact verification to reduce fake news. Section 3 determines our research methodology. Section 4 discusses the challenges of fact verification and the future scope of fact verification. Section 5 discusses the conclusions of the literature study conducted.

II. Literature Review

Fact verification is a primary need to deal with the widespread spread of false information, especially in Indonesia [27]. Researchers have undertaken a series of challenges to provide innovative solutions for fact verification, one of which is creating a series of data collections in the form of datasets that can be used for ongoing research. Currently, fact verification datasets are mostly intended for English [14], [20], [21]. Handling fake news by creating an Indonesian language dataset has actually been published [28]. Pratiwi et al. used a machine learning algorithm to identify hoax news in Indonesian, marking a positive step in combating fake news. However, their dataset does not provide evidence to verify whether the news is true or false. Including evidence would enhance information confidence for news consumers. Significant breakthroughs in the field of fact verification were made through the development of English language datasets with evidence, such as FEVEROUS, EX-FEVER, and FACTIFY-5WQA. These datasets use machine learning and NLP techniques to test the veracity of claims. FEVEROUS, for example, provides claims and supporting evidence, adding to the complexity of fact-checking. Although effective in English, application in Indonesian is still limited and requires NLP adjustments [29], [30], due to linguistic differences. For dataset effectiveness, researchers use models such as RoBERTa or XL-NET. Other studies, such as FACTKG [14], also expanded the dataset. Transformer algorithms, part of deep learning [12], [31], have successfully understood and processed textual data with high efficiency.

Furthermore, several deep learning algorithms have been specifically developed for fact verification tasks, such as Unified Evidence Extraction for Fact Verification (UNIFEE) [32], Input-Oriented Demonstration Learning (IODL) [33], Heuristic Heterogeneous Graph Reasoning Networks (H2GRN) [34],and Generative Evidence Retrieval (GERE) [35]. All of these algorithms focus on the application of neural networks. Related research suggests several improvements, including enhancing the efficiency and sensitivity of the models by using a single decoder. Additionally, an end-to-end approach is recommended to improve the quality of the generated evidence while being more efficient in the computation time for fact verification processing [36]. Large Language Models (LLMs), deep learning models based on transformers [37], can also be applied in the context of single decoders for text processing. Some of the popular LLMs used for fact verification include Bidirectional Encoder Representations from Transformers (BERT) [26], A Robustly Optimized BERT Pretraining Approach (RoBERTa) [38], Cross-lingual Language Model (XLM) [39], and Generative Pre-trained Transformer (GPT) [25]. LLMs use transformer architecture to process and understand linguistic context in text by predicting the next words that will appear in a sentence. Their ability to capture relationships between sentences and words in text has been demonstrated in fact verification studies [40]. Finally, a comparison of fact verification research looking at the systematic literature can be seen in Table I.

TABLE I. Comparison Studies

Criteria	This Study	[41]	[42]	[43]
The paper is an SLR that focuses on fact verification	Yes	Yes	No	No
Presents framework for preparing literature reviews	Yes	No	No	No
Presents information on the latest issues regarding the spread of fact verification to detect fake news	Yes	No	Yes	Yes
Presents model performance information in national languages	Yes	No	Yes	No
Presents information on fact verification methods to detect fake news	Yes	Yes	No	Yes
Presents the dataset that is often used to fact verification to detect fake news.	Yes	Yes	Yes	Yes
Presents information on technological developments carried out in carrying out fact verification to detect fake news	Yes	Yes	Yes	Yes
Presents information on how to get good performance on the model in carrying out fact verification to detect fake news	Yes	Yes	Yes	Yes

III. Methodology

This study uses a Systematic Literature Review (SLR) methodology to assess all research on fact verification. The

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methodology includes five phases: planning, literature search, setting inclusion and exclusion criteria, data collection, and data analysis. Fig. 2 gives an overview of this Fact Verification SLR research, highlighting contemporary facets and technological concerns in fact verification.

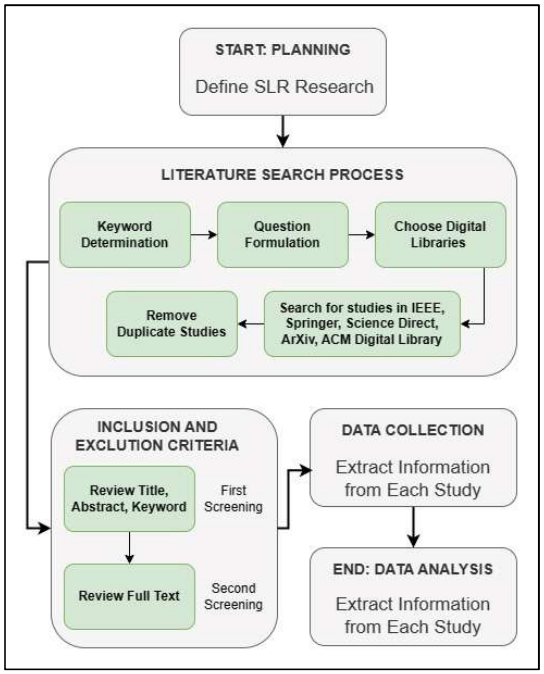


Fig. 2. Overview of Fact Verification SLR Research

In Fig. 2, the initial phase, planning, outlines the strategies to be implemented throughout the research endeavor. The subsequent phase, the literature search process, involves the compilation of pertinent information pertaining to fact verification. The third phase, delineating inclusion and exclusion criteria, ensures the incorporation of only pertinent studies into the analysis. The fourth phase, data collection, entails the extraction of pertinent data from the selected studies. Finaly, the data analysis phase is employed to assess the most recent trends and technological advancements within the scope of fact verification. Overall, this methodology furnishes a comprehensive comprehension of the subject matter and facilitates a review of recent progressions and challenges encountered within the domain of fact verification.

A. Define SLR Research

Determining SLR research that aims to classify fake news begins by defining Population, Intervention, Comparison, Outcomes, and Context (PICOC). PICOC is structured to direct the research focus in forming a pattern of information to be utilized for the research. Table II provides a summary of research definitions using the PICOC structure and the formation of keywords to be used during the literature review search process.

TABLE II. COMPARISON STUDIES

NO	PICOC	Definition	Keyword
1	Population	Literature on the subject of fact verification.	Fact Verification
2	Intervention	Literature that discusses the classification and prediction of fake news or fact verification.	Method, Model, Algorithm
3	Comparison	Datasets and Models.	Dataset, FEVER, FEVEROUS
4	Outcomes	Literature shows the success of research as measured by Dataset Score, Accuracy, Recall, Precision, F1-Score.	-
5	Context	Journal published maximum in 2020.	-

B. Literature Research Process

The literature search process is determined based on the keywords that have been arranged in Table II, and question formulations are formed during the literature search process. The question formulation is developed as we review the articles. After conducting the literature review, the formulation of questions was summarized in Table III.

TABLE III. FORMULATED RESEARCH QUESTION

NO	Question	Origin and Context
Q1	What issues are frequently discussed in fact verification?	Based on the literature, most fact verification research addresses issues generally, without being tied to specific topics. This question arises to explore specific issues and address research gaps in fact verification.
Q2	What are the labels for the classification results?	This question aims to understand the classification standards used to identify classification patterns in fact verification.
Q3	What dataset is used for fact verification?	The literature shows that most fact verification datasets are derived from English sources. This question seeks to explore whether other datasets have been developed, particularly for Indonesian.
Q4	Is Indonesian applied to fact verification datasets?	This question arises to evaluate the application of Indonesian in fact verification datasets, considering that its use is still very limited compared to English.
Q5	What algorithm are developed to fact verification?	This question aims to highlight the development of algorithms for fact verification in a multi-language context.
Q6	How does the model recognize entities in text in carrying out classification?	This question explores how Natural Language Processing (NLP)-based models recognize entities in text and classify them, which is crucial for determining the accuracy of fact verification, especially in claims with specific context.
Q7	What Indicators are used to measure algorithm performance?	This question seeks to understand the indicators used to measure the performance of fact verification algorithms, such as accuracy, precision, recall, and F1-Score, and their application in the Indonesian context.

The next step is to select relevant literature to conduct a literature study. The libraries used to create this literature study are IEEE Xplore (12 articles), ScienceDirect (21 articles), ACM Digital Library (8 articles), and ArXiv (148 articles). Once the data is collected from multiple libraries, duplicate titles are removed before sorting the titles based on fact verification to detect fake news.

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C. Equations Inclusion and Exclusion Criteria

At this stage, the researcher reviews the title, abstract, and keywords of the study that has been compiled to determine whether it is relevant to the research topic. This process filters out inappropriate studies without the need to read the entire text of the document. The inclusion and exclusion stages carried out are summarized in Table IV.

TABLE IV. INCLUSION AND EXCLUSION CRITERIA

Inclusion	Exclusion
The study discusses the dataset used in combination with a fact verification algorithm.	The study not discusses the dataset used in combination with a fact verification algorithm.
The study discusses fact verification algorithms to detect fake news.	The study not discusses fact verification algorithms to detect fake news.
The studies discuss measuring the performance of datasets or algorithms.	The studies not discuss measuring the performance of datasets or algorithms.
The maximum year of scientific article publication is the last 4 years.	The year of publication of scientific articles is more than 4 years.

The selection stage is conducted in two phases: the first involves selecting the title, abstract, and keywords. Initially, the articles collected from the predetermined libraries totaled 199 but were reduced to 60 articles (12 from IEEE, 12 from ScienceDirect, 8 from ACM Digital Library, and 30 from ArXiv) due to the year not meeting the criteria. Furthermore, after reviewing the full text of all the articles, the final result was 35 articles (12 from IEEE, 6 from ScienceDirect (SDr), 2 from ACM Digital Library (ACMDL),10 from ArXiv, and 6 from another library) suitable for comparison in the literature study.

D. Data Collection

The data collection carried out at this stage involves gathering information, specifically collecting data related to the prepared questions and then compiling the data to be used as a source of information for developing datasets and fact verification models. The mapping of the literature results to the data collection can be seen in Table V, this table shows the alignment of each research article with the prepared research questions, providing a clear overview of how each article contributes to the development of datasets and fact verification models.

TABLE V. MAPPING OF ARTICLES TO RESEARCH QUESTIONS

NO	Paper	Origin	Q1	Q2	Q3	Q4	Q5	Q6	Q7
1	[16]	ArXiV	✗	✓	✓	✗	✓	✓	✓
2	[19]	ArXiV	✗	✓	✓	✗	✗	✓	✓
3	[20]	ArXiV	✗	✓	✓	✗	✗	✓	✓
4	[21]	ArXiV	✗	✓	✓	✗	✗	✗	✓
5	[25]	ArXiV	✗	✓	✓	✗	✓	✗	✓
6	[26]	ArXiV	✗	✓	✓	✗	✗	✗	✓
7	[44]	IEEE	✓	✗	✓	✗	✓	✗	✓
8	[45]	IEEE	✗	✓	✓	✗	✓	✓	✓
9	[46]	IEEE	✗	✓	✓	✗	✓	✓	✓
10	[47]	IEEE	✗	✓	✓	✗	✓	✓	✓
11	[48]	IEEE	✗	✓	✓	✗	✓	✓	✓
12	[33]	SDr	✗	✓	✓	✗	✓	✓	✓
13	[49]	ArXiV	✗	✓	✓	✗	✓	✓	✓
14	[32]	ACL	✗	✓	✓	✗	✓	✓	✓
15	[50]	ACMDL	✗	✓	✓	✗	✓	✓	✓
16	[51]	ArXiV	✗	✓	✓	✓	✓	✗	✓
17	[52]	SDr	✗	✓	✓	✗	✓	✓	✓
18	[53]	IEEE	✗	✓	✓	✗	✓	✗	✓
19	[54]	SDr	✗	✓	✓	✗	✓	✗	✓
20	[34]	IEEE	✗	✓	✓	✗	✓	✗	✓
21	[35]	ACMDL	✗	✓	✓	✗	✓	✗	✓
22	[55]	SDr	✗	✗	✗	✗	✓	✗	✓
23	[56]	IEEE	✗	✓	✓	✗	✗	✗	✓
24	[57]	SDr	✗	✓	✗	✗	✗	✗	✓
25	[58]	Stanford	✗	✓	✓	✗	✓	✗	✓
26	[59]	ArXiV	✗	✓	✓	✗	✓	✗	✓
27	[40]	AAAI	✗	✓	✓	✗	✓	✗	✓
28	[60]	ArXiV	✗	✓	✓	✗	✓	✗	✓
29	[61]	IJSI	✗	✓	✓	✗	✓	✗	✓
30	[62]	IEEE	✗	✓	✓	✗	✓	✗	✓
31	[63]	PLoS ONE	✗	✓	✓	✗	✓	✗	✓
32	[64]	IEEE	✓	✓	✗	✗	✓	✗	✓
33	[65]	IEEE	✗	✓	✗	✗	✓	✗	✓
34	[66]	IJETT	✗	✓	✗	✗	✓	✗	✓
35	[67]	SDr	✗	✓	✗	✗	✓	✗	✓

E. Data Analysis

Furthermore, after the data has been extracted, the information is processed for data analysis. At this stage, patterns from the datasets and models used in previous studies are further analyzed using descriptive and comparative statistical techniques. This analysis aims to delve deeper into the methods used for creating and implementing datasets and fact verification models, identifying performance trends and challenges, and providing a foundation for the development of more effective models in the future.

IV. RESULT AND DISCUSSION

Analysis of literature studies shows challenges in developing fact verification datasets in Indonesian. The

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existing datasets are limited to certain languages, making it difficult to adapt them to Indonesian. From the research summarized in this subsection, challenges emerge in developing datasets and models for future research.

A. *Q1: What issues are frequently discussed in fact verification?*

After being traced to the literature conducted, in general, research related to fact verification does not discuss specific topics that are used in the purpose of research to deal with fake news. Only research conducted by Ishikawa et al., [44] which discusses the topic specifically, which discusses the verification of facts on the topic of Politics. The research depends on human annotators to create false claims for test data, which can impact the consistency and objectivity of the data, especially without strict standards in the process.

B. *Q2: What are the labels for the classification results?*

The articles we reviewed mostly classify results as Supported and Refuted [45], [46], [47], [48]. Supported and Refuted classifications come with evidence, while claims lacking evidence are categorized as Not Enough Info. A drawback is that much of the evidence provided lacks metadata, such as links to information sources. This presents an opportunity to develop Indonesian fact verification datasets that include such metadata.

C. *Q3: What dataset is used for fact verification?*

Some studies use more than 1 dataset to compare algorithm performance. From 35 studies, we found that there were 5 uses of datasets and the dominant one used was the FEVEROUS [16], [32], [33], [49], [50]. FEVEROUS appears six times in the number of datasets used. Next, we reviewed the FEVEROUS dataset in the main FEVEROUS article [19]. FEVEROUS, a dataset developed from FEVER, contains 87,000 claims combined with structured and unstructured data. Fig. 3 is an example of the FEVEROUS dataset used for fact verification.

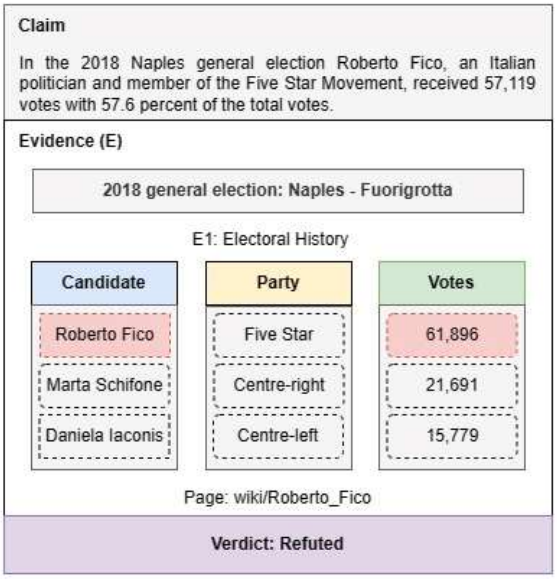


Fig. 3. Example of a claim from FEVEROUS Dataset

The claim states that in the 2018 Naples general election, Roberto Fico received 57,119 votes with 57.6 percent of the total votes. FEVEROUS does not specifically categorize topics in its dataset, which is sourced from Wikipedia, and it does not include political topics related to world wars and similar events. Fig. 3 is an overview of FEVEROUS. However, evidence from Wikipedia shows that Roberto Fico actually received 61,896 votes. Therefore, the claim is refuted because it does not match the available data. FEVEROUS uses structured and unstructured data from Wikipedia to provide accurate evidence in fact verification tasks. However, its reliance on human annotators affects data consistency and objectivity. Additionally, FEVEROUS is limited to English, and similar architectures have not been tested in other languages combined with models.

D. *Q4: Is Indonesian applied to fact verification datasets?*

Several studies used multiple languages in their datasets, with 30 studies employing English in the training and testing processes. Only one study was found that used Indonesian, conducted by Y.C. Chang et al., [51]. In that study, XFEVER utilizes machine translation to build datasets across various languages, including Indonesian. The findings suggest that zero-shot methods, trained solely on English data, may struggle with significant linguistic differences in Indonesian. This is evident from the difference in accuracy between English and Indonesian: for example, in the mBERT model, the accuracy for English reached 87.9%, while for Indonesian it was 82.6%, indicating a 5.3% decrease. Similarly, in the XLM-R-large model, English achieved 89.5% accuracy, while Indonesian had 85.5%, showing a 4.0% gap. While machine translation offers a feasible solution, it introduces contextual errors that may impact the accuracy of fact verification models trained on XFEVER data. These challenges highlight the complexity of developing a robust fact verification dataset in Indonesian, which requires further refinement of both translation techniques and model adjustments.

E. *Q5: What algorithm are developed to fact verification?*

Models that are developed specifically for fact verification generally face a variety of limitations related to their performance. One of the key challenges is that the performance of these models often falls below the threshold required to gain human confidence, which is typically set at an accuracy rate of 70% or higher [68], [69]. This means that when the accuracy of the model is below 70%, it is considered insufficiently reliable for users to trust its outputs. To provide a clearer understanding of how different models perform in this domain, we have conducted a comprehensive comparison and the results of this comparison are presented in Table VI. These results highlight the strengths and weaknesses of various models designed specifically for fact verification.

TABLE VI. COMPARISON OF FACT VERIFICATION MODELS

Model	Precision > 70%	Recall >70%	F1-Score > 70%	Accuracy >70%
UNIFEE [35]	No	No	No	Yes
H2GRN [34]	No	No	No	Yes
GERE [35]	No	Yes	No	Yes
RERG [54]	No	No	No	Yes

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We highlighted that one of the algorithms that carried out the comparison, i.e., the GERE algorithm [35], has a score threshold above 70% for the Recall and Accuracy measurement indicators. As we look deeper into the architecture of GERE, the algorithm has limitations in terms of independent scores that fail to capture interactions. The generative approach used by GERE tends to separate the process of assessing the relevance of documents and sentences, which results in losing local context and interactions between documents and sentences. This can reduce the effectiveness of fact verification when claims require analysis of interrelated evidence from multiple sources. This failure is evidenced by the high recall results and lower precision, indicating that this model fails to retrieve all relevant evidence that may be needed for comprehensive claim verification. The development of transformer models for fact verification, particularly for the Indonesian language, has significant potential compared to neural network-based models. Transformer algorithms such as BERT [55], RoBERTa [56], XLM [57], and GPT [58] are focused on the ability to capture interactions between documents and sentences. This statement is supported by research that shows transformer algorithms have good sensitivity in performing fact verification tasks.

F. *Q6: How does the model recognize entities in text in carrying out classification?*

Based on the results of the literature study conducted, we found that the majority of studies used Natural Language Processing (NLP) [39], [40], [59], [60], [61], [62], [63] which had an impact on the quality of the model in carrying out fact verification tasks. In Fig. 4, we summarize our findings that NLP is one of the methods most commonly used each year to help models recognize entities.

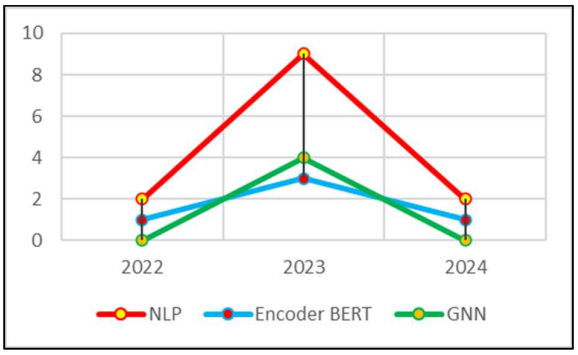


Fig. 4. NLP is widely used to recognize text entities

We conclude that the proper application of NLP impacts a model's ability to understand context and language. Another method that emerges from our findings is the application of the BERT Encoder and Graph Neural Network (GNN) to understand the language context of the model. However, these two methods are not specifically designed to understand linguistics. We conclude that NLP methods could present a new challenge in helping models understand Indonesian in datasets.

G. *Q7: What Indicators are used to measure algorithm performance?*

Accuracy, precision, recall, and F1-Score are general indicators used to measure model performance based on the results of our literatre review [64], [65], [66], [67]. Additionally, other measurements used to assess the quality of a dataset include the Dataset Score, which is also widely applied in the research we reviewed. These indicators can serve as references for future research to measure the quality of datasets and models in carrying out fact verification, especially in Indonesian. Fig. 5 is a visualization of the number of studies measuring model performance using indicators such as accuracy, precision, recall, F1-Score, and other measurements (Support [62] and AUC [63]).

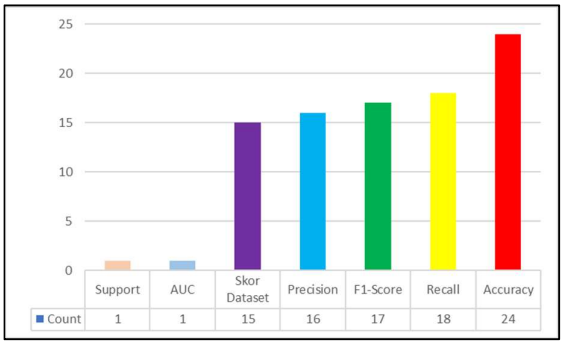


Fig. 5. Most Used Performance Measurement in Fact Verification Research

H. *Discussion*

The discussion identified several major shortcomings in existing research on fact verification in general. One prominent issue is the low performance of current models, which often fail to achieve the desired accuracy and reliability, generally falling below the 70% threshold required to gain user trust. This low performance is mainly due to the models' inability to handle the linguistic and contextual challenges of natural language processing. In particular, the lack of dedicated datasets designed for fact verification in Indonesian is a significant obstacle. Although there is one study that tested Indonesian, the language was not involved in the training process in Natural Language Processing (NLP). This gap represents an opportunity for fact verification research in the Indonesian language context, aiming for good model performance with accuracy, precision, recall, and F1-Score all reaching values above 70%. Additionally, the discussion could be more in-depth by addressing political topics in Indonesia, as these are rarely discussed in fact verification cases despite the widespread prevalence of fake news related to politics in the country. Overall, this discussion highlights the need for improved datasets and more sophisticated models adapted to Indonesian, with a deeper focus on political cases. Future research should focus on developing datasets that include metadata and contextual information, as well as improving model architectures to capture complex interactions between documents and sentences. Furthermore, expanding research into multiple domains can provide valuable insights and improve the overall reliability and effectiveness of fact verification systems.

V. Conclusion

In this study, we review literature on the development of datasets and fact verification models with the main aim of detecting fake news using fact verification methods. The findings highlight the importance of developing datasets and algorithms specifically for Indonesian by adapting the fact verification process to the language. Despite major advances in English-based fact verification systems, the unique challenges that arise in the Indonesian language context

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require tailored solutions. Based on our research objectives, we conclude that:

- Establish collaborations with local media and fact verification organizations to create and metadata large-scale Indonesian datasets.
- Enhance model architectures to improve performance metrics, with a target accuracy above 70%.
- Conduct cross-language studies to adapt successful models and datasets from other languages to the Indonesian context.

The journey towards reliable fact verification in Indonesia involves ongoing development and evaluation of datasets and models designed specifically for the Indonesian context. By addressing this gap, future research can improve the reliability and efficiency of fact verification systems, which will benefit the growth of a more informed and knowledgeable digital society in Indonesia. Additionally, these recommendations have the potential to influence advancements in machine learning and natural language processing (NLP), particularly in low-resource languages. By addressing the technical challenges of adapting models for fact verification in Indonesian, this research could contribute to the refinement of algorithms, the development of datasets in Indonesia, particularly in the context of political news, and further advance the state of Artificial Intelligence (AI) across a wide range of applications.

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